



Data Structures and Algorithms -3 (25CS2103E)

Symptoms-to-Diagnosis Matcher

Under the Esteemed Guidance of

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Abstract

- Java-based system that matches patient symptoms with possible diagnoses.
- Normalizes symptoms and compares them with a medical knowledge base.
- Scores and ranks candidate diagnoses.
- Recommends minimal diagnostic tests to distinguish conditions.
- Uses DSA concepts like HashMap, HashSet, ArrayList, and PriorityQueue.

Introduction

- Medical symptoms can be described in different ways by users.
- This makes direct matching with a medical knowledge base difficult.
- Our system normalizes symptoms into standard terms.
- It then matches, scores, and ranks possible conditions.
- DSA concepts help make the system efficient and explainable.

Literature Survey

- Clinical decision-support systems help in diagnosis and medical decision-making.
- Existing diagnostic systems can reduce the time needed to identify possible conditions.
- Research shows that interpretability and transparency are important for healthcare decision-support systems.
- Recent studies highlight challenges such as accuracy, usability, clinical integration, and trust.
- Our project focuses on transparent symptom matching, diagnosis ranking, and minimal test selection



Problem Statement and Objectives

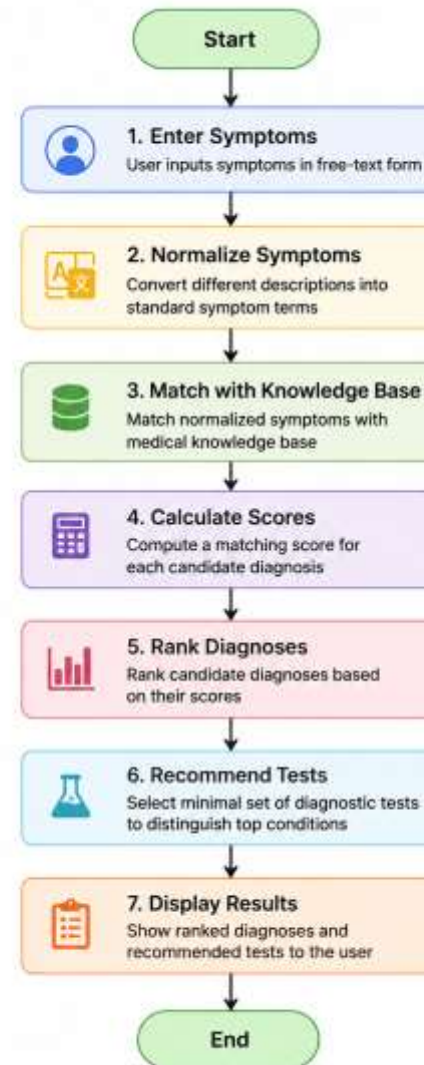
Problem Statement

- To develop a Java-based system that **normalizes patient symptoms, matches them with possible medical conditions, ranks diagnoses using transparent scoring, and recommends minimal diagnostic tests** to distinguish between the top conditions.

Objectives

- **Normalize** free-text symptoms into standard terms.
- **Match** symptoms with the medical knowledge base.
- **Score** candidate diagnoses based on symptom matches.
- **Rank** the most relevant diagnoses.
- **Recommend** minimal tests to distinguish between conditions.
- Apply **DSA concepts** such as hashing, searching, sorting, and ranking.

Flow Chart



Results

- Successfully **normalizes** free-text symptoms into standard terms.
- Matches symptoms with possible **candidate diagnoses**.
- Calculates a **transparent score** for each diagnosis.
- Ranks the most relevant conditions based on their scores.
- Recommends a **minimal set of diagnostic tests** to distinguish top conditions.
- Demonstrates the use of **DSA concepts** such as hashing, searching, sorting, scoring, and ranking.

Hardware and Software requirements

Hardware

- **Processor:** Intel Core i3 or equivalent
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 500 MB
- **Display:** Standard monitor
- **Input:** Keyboard & mouse
- **Internet:** Required for development tools/resources

Software

- **OS:** Windows
- **Language:** Java
- **JDK:** 17 or above
- **IDE:** Visual Studio Code
- **Database:** MySQL
- **Frontend:** Java Swing
- **Version Control:** GitHub

Conclusion

- The system matches **patient symptoms with possible diagnoses**.
- It uses **normalization, scoring, and ranking** for transparent results.
- It recommends **minimal diagnostic tests** to distinguish top conditions.
- DSA concepts make the system **efficient and explainable**.
- The project demonstrates a practical application of **Data Structures and Algorithms** in healthcare.

Future Scope

- Expand the **medical knowledge base** with more symptoms and conditions.
- Improve **symptom normalization** for synonyms and different descriptions.
- Enhance **diagnosis scoring and ranking**.
- Improve the **diagnostic test selection** process.
- Add more patient information such as **symptom duration and severity**.
- Develop **web and mobile versions** of the system.
- Add **multilingual symptom support**.
- Improve the **explainability** of diagnosis scores and recommendations.

Thank you